

RUTHERFORD COUNTY AP, NC, USA

WMO: 723144

Lat: 35.428N

Lon: 81.935W

Elev: 329

StdP: 97.44

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53890

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.9	-4.4	-19.0	0.7	1.3	-16.3	0.9	1.5	8.3	8.8	7.3	8.1	0.8	350	0.354

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	33.8	22.7	32.5	22.9	31.2	22.5	25.1	29.8	24.5	29.0	24.0	28.3	1.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.5	26.5	23.0	18.5	26.0	22.7	18.1	25.7	78.3	29.6	75.7	28.8	73.7	28.4	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.9	5.5	4.7	-11.2	34.6	2.8	1.8	-13.2	35.9	-14.9	36.9	-16.4	37.9	-18.5	39.2		
			-12.0	26.1	2.3	0.7	-13.7	26.6	-15.1	27.0	-16.4	27.4	-18.1	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.7	5.2	7.1	10.6	15.6	19.7	23.8	25.2	24.8	21.9	16.1	10.7	6.8
	DBStd	8.15	5.43	4.74	4.82	4.16	3.29	2.33	2.26	2.12	2.99	4.15	4.44	4.71
	HDD10.0	504	169	104	53	6	0	0	0	0	0	5	43	123
	HDD18.3	1782	408	316	244	101	26	1	0	0	8	91	230	359
	CDD10.0	2572	19	23	71	174	302	415	471	460	356	194	64	23
	CDD18.3	809	0	1	3	18	69	166	213	202	114	22	2	1
	CDH23.3	6875	2	4	43	233	607	1431	1869	1601	866	196	19	4
	CDH26.7	2380	0	0	5	39	149	522	766	591	277	31	1	1
Wind	WSAvg	1.5	1.6	1.7	1.9	2.1	1.6	1.3	1.1	1.0	1.3	1.4	1.3	1.3
Precipitation	PrecAvg	1338	123	106	136	114	94	119	124	121	118	89	91	101
	PrecMax	1762	209	189	210	243	226	222	250	214	378	301	226	174
	PrecMin	965	45	43	49	34	48	24	61	16	29	0	30	46
	PrecStd	229	52	42	47	61	46	64	46	60	82	78	48	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	23.0	27.3	29.8	32.1	34.2	36.2	34.5	33.0	29.2	26.1	22.5
		MCWB	14.3	15.5	15.9	18.1	20.8	22.1	23.0	22.7	21.3	20.6	16.3	15.3
	2%	DB	18.9	20.3	24.1	27.6	29.9	32.8	33.9	32.9	31.8	27.3	22.9	19.2
		MCWB	13.6	13.2	14.3	16.7	20.1	22.7	23.0	23.7	21.1	18.5	15.1	15.2
	5%	DB	17.0	17.7	22.2	26.2	28.0	31.3	32.5	31.6	29.8	26.0	21.3	17.3
		MCWB	12.5	11.7	13.7	16.2	19.5	22.4	23.5	23.1	21.4	17.8	14.5	12.2
	10%	DB	14.0	16.0	19.2	24.1	27.1	29.4	31.1	29.9	27.9	23.6	18.8	14.8
		MCWB	9.6	10.4	12.2	15.5	19.1	21.8	23.2	22.5	20.7	17.4	13.7	11.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	17.9	17.8	20.6	23.0	25.1	25.8	25.8	24.1	23.3	20.5	19.0
		MCDB	19.7	20.7	23.1	25.1	26.8	29.6	30.9	30.6	27.9	25.7	22.5	20.8
	2%	WB	15.4	16.0	16.4	19.3	22.0	24.2	25.1	25.1	23.4	21.4	17.9	16.2
		MCDB	18.0	18.4	21.2	24.2	26.2	29.2	30.0	29.7	27.1	24.5	20.1	18.2
	5%	WB	13.6	13.6	15.2	18.3	21.4	23.6	24.5	24.5	22.6	20.0	16.5	14.0
		MCDB	16.1	16.5	19.7	23.1	25.7	28.5	29.3	28.7	26.4	23.7	19.0	15.9
	10%	WB	10.9	11.3	13.8	17.2	20.6	23.0	24.0	24.0	22.0	18.6	14.9	11.7
		MCDB	12.8	14.2	17.9	21.5	24.8	27.7	28.5	27.8	25.8	22.3	18.1	13.7
Mean Daily Temperature Range	5% DB	MDBR	11.1	11.5	12.2	12.7	11.9	11.0	10.5	9.8	10.5	12.3	12.7	10.8
		MCDBR	13.7	14.9	15.8	15.7	13.9	13.5	13.0	12.2	13.4	14.1	15.6	13.8
		MCWBR	8.8	9.1	8.1	7.1	5.9	5.0	4.6	4.4	4.7	6.7	8.5	8.8
	5% WB	MCDBR	10.8	11.9	11.9	12.3	11.1	10.8	10.9	9.8	9.9	10.4	10.7	10.6
		MCWBR	8.0	8.9	7.4	6.6	5.5	4.8	4.7	4.2	4.6	6.2	7.6	8.4
Clear-Sky Solar Irradiance	taub	0.304	0.313	0.345	0.393	0.446	0.475	0.520	0.512	0.409	0.366	0.323	0.304	
	taud	2.524	2.492	2.435	2.300	2.177	2.134	2.037	2.039	2.312	2.420	2.503	2.554	
	Ebn at Noon	903	934	925	892	844	817	777	775	847	859	871	883	
	Edh at Noon	81	93	107	129	148	155	170	166	120	99	82	74	
All-Sky Solar Radiation	RadAvg	2.61	3.26	4.30	5.33	5.90	6.24	5.81	5.39	4.62	3.84	2.90	2.24	
	RadStd	0.15	0.36	0.29	0.41	0.46	0.44	0.40	0.26	0.42	0.43	0.25	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+165	+106

Nomenclature: See separate page